Quantifying the accuracy of quantiles

Statistics and Data Analysis

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1 Task

Consider the gamma distribution with shape parameter equal 3 and rate 0.5. We are interested in the accuracy of quantile estimation from a sample of $n = 50$ observations from that distribution. Consider quantiles corresponding to the following probabilities $\{0, 0.2, 0.4, 0.6, 0.8, 0.95, 0.99, 1\}$.

- Generate the sample of n observations from that distribution, compute empirical quantiles and compare them with the actual quantiles of the underlying distribution
- Use bootstrap to compute 95% confidence intervals for each of the empirical quantiles.
- Plot the underlying quantiles, empirical quantiles and confidence intervals against the probabilities.
- Do you notice any biases? How does the length of the confidence interval depend on the probability?

2 Tips

- Quantiles of the underlying distribution can be generated using `qgamma(p = probs, shape = 3, rate = 0.5)`